FinSight

ARIN7012 Project: Evidence-First Financial Analysis Chatbot

Group 4.2

The University of Hong Kong

ARIN7012 Project Presentation

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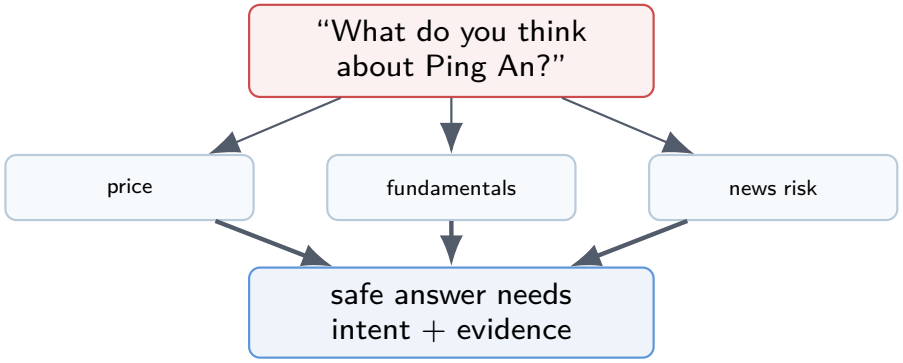
Background and Goal

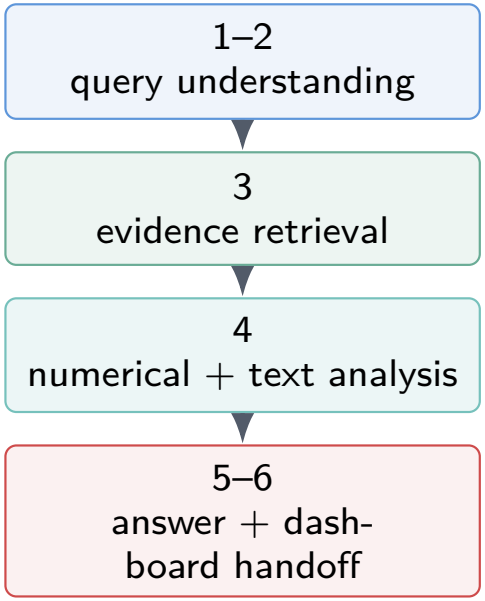
Financial chatbot answers must be grounded

- Finance questions may affect buy, hold, or sell decisions.
- User intent is often hidden inside short natural language.
- Market data, news, and macro signals change quickly.
- Answers need traceable evidence and clear limitations.

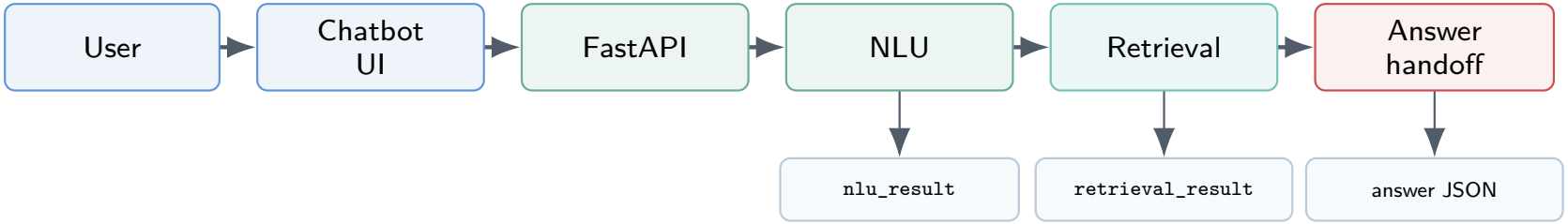


The project goal is an evidence-first financial chatbot

- Understand product type, entities, intent, topic, and risk.
- Search likely answer sources: price, news, fundamentals, macro.
- Analyze numerical and text evidence before response generation.
- Produce answer JSON and next-question suggestions for the chatbot.



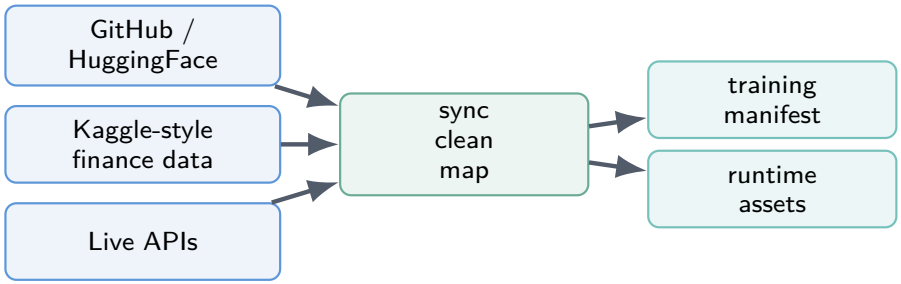
FinSight is powered by the ARIN intelligence backend



The frontend is the user-facing shell. This deck focuses on the backend intelligence and evidence contract.

Data flows from public sources to runtime evidence

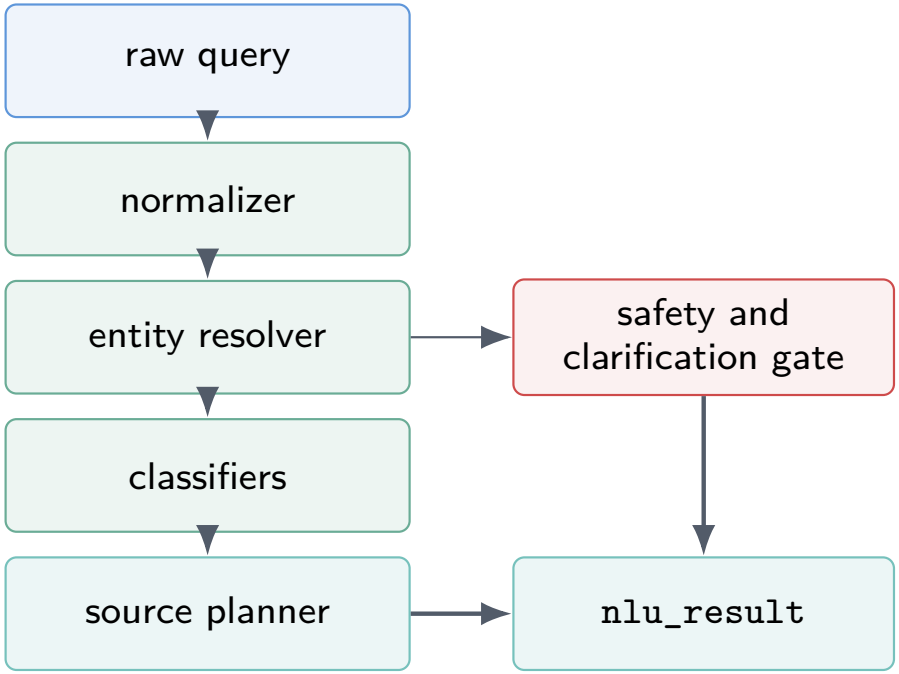
- Public datasets support training and evaluation.
- Runtime entities, aliases, and documents support clone usage.
- Live providers add fresh market, news, announcement, and macro data.
- Cleaning maps raw data into task assets and searchable evidence.



Implementation and Performance

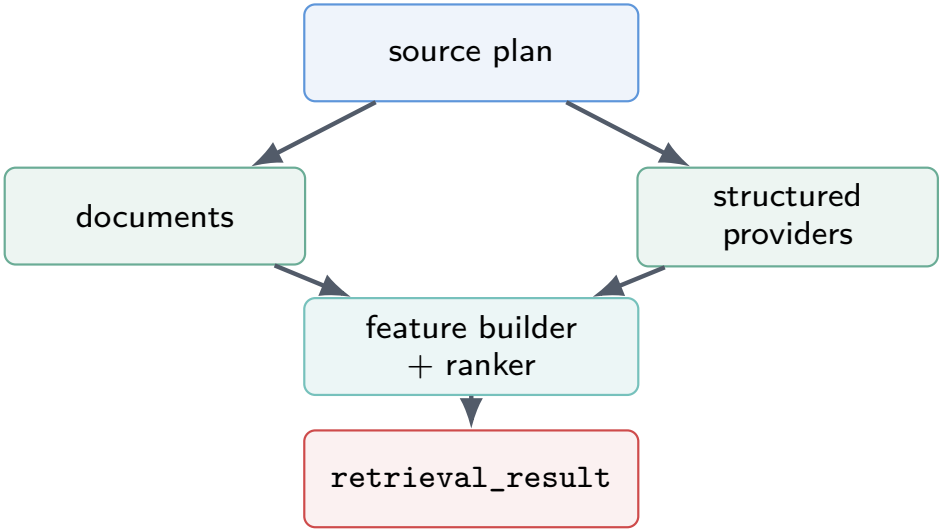
NLU turns raw language into source requirements

- Normalizer handles Chinese and English query variants.
- Resolver links aliases, fuzzy matches, CRF spans, and typos.
- Classifiers predict product, intent, topic, and question style.
- Output includes missing slots, risk flags, and `source_plan`.



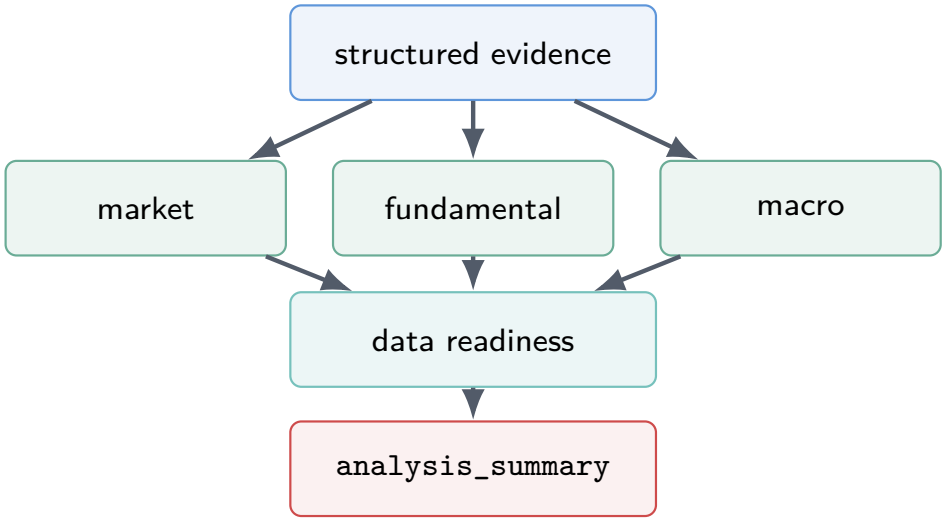
Retrieval gathers ranked evidence from documents and APIs

- Query builder expands the NLU source plan.
- Document retriever returns news, announcements, reports, FAQ, and product docs.
- Structured providers return market, fundamental, industry, and macro rows.
- Ranker and packager output coverage, warnings, and trace IDs.



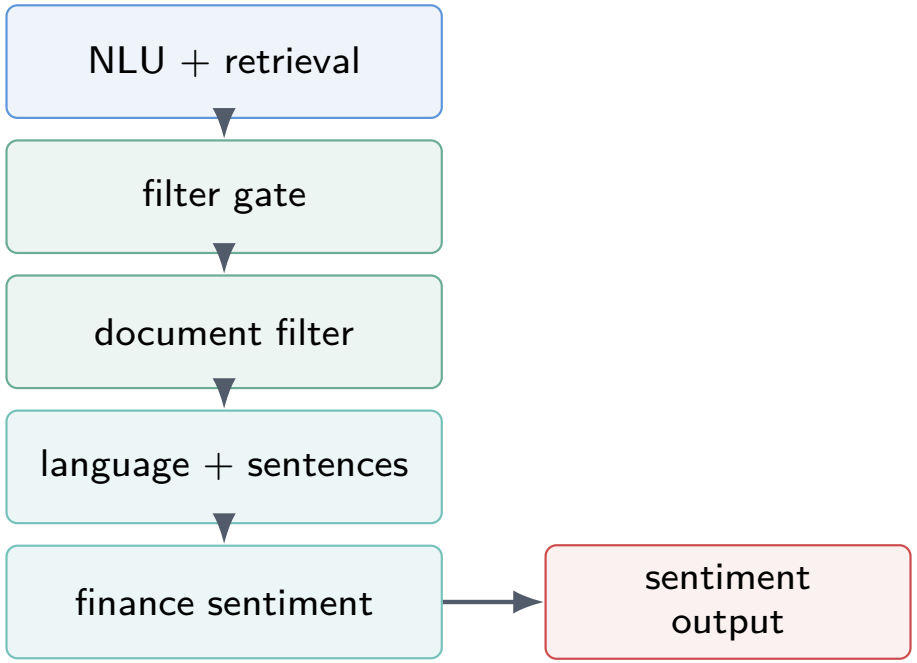
Market analysis summarizes signals without giving advice

- Market signal: returns, MA5/MA20, RSI, MACD, volatility.
- Fundamental signal: PE, PB, ROE, and valuation assessment.
- Macro signal: available indicators and direction.
- Data readiness shows which evidence categories are complete.



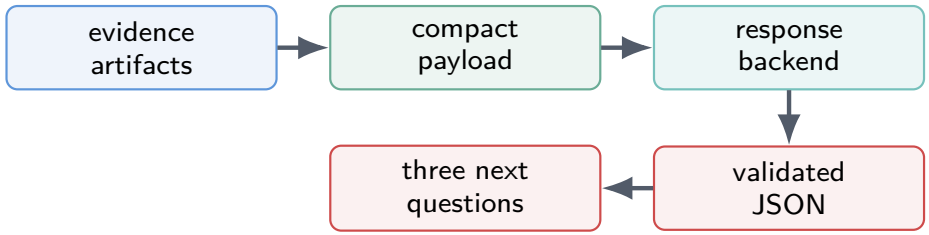
Sentiment analysis reuses retrieved evidence and entity context

- Runs after NLU and retrieval, not as a second understanding system.
- Skips out-of-scope, product-info, fee, and missing-entity cases.
- Filters documents by source type, language, and entity relevance.
- Emits document-level `SentimentItem` records.



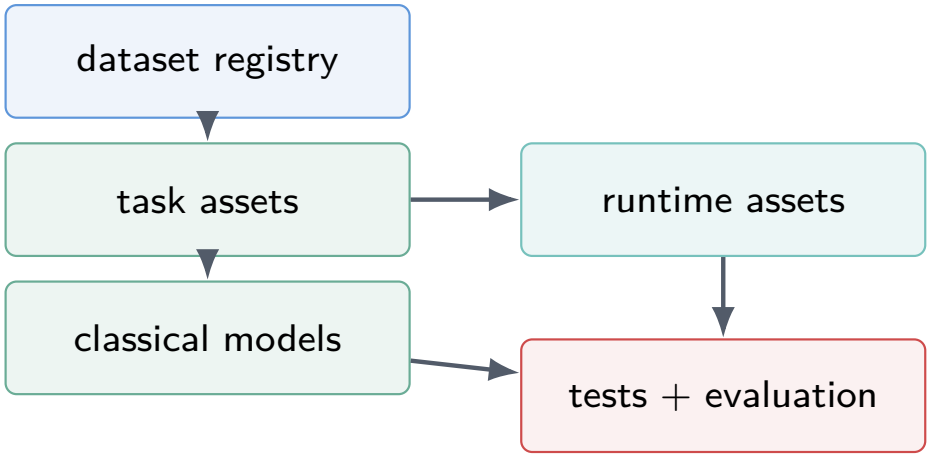
Answer handoff gives the chatbot validated JSON

- Compacts evidence before sending it to the response backend.
- Strips raw debug traces and long documents.
- Validates `answer_generation` and `next_question_prediction`.
- Keeps evidence IDs, limitations, and a risk disclaimer.



Training and evaluation keep the backend measurable

- Core NLU and retrieval use classical, explainable ML.
- Dataset sync builds manifest-based task assets.
- Runtime assets are separated from training caches.
- Tests cover JSON schemas, boundaries, fuzz cases, and integrations.



Current results support the demo story

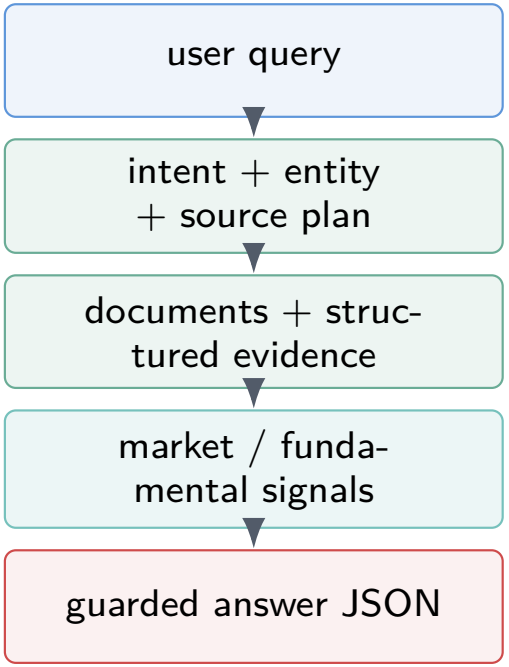
0.9881	0.9750	0.8727	0.9474	0.8849
Finance recall	OOD rejection	Intent F1	Topic F1	NDCG@10
10k eval	10k eval	10k eval	10k eval	retrieval

Check	Result
Large evaluation	10,000 bilingual finance and out-of-domain queries; all thresholds passed.
Fuzz matrix	32/32 cases passed; 361/361 checks passed.
Focused test run	193 passed, 41 skipped.

Demo and Conclusion

Demo: one query becomes four evidence layers

- Example: "What do you think about Ping An Insurance?"
- The chatbot sends raw text and optional user profile.
- FinSight returns structured artifacts for response generation.

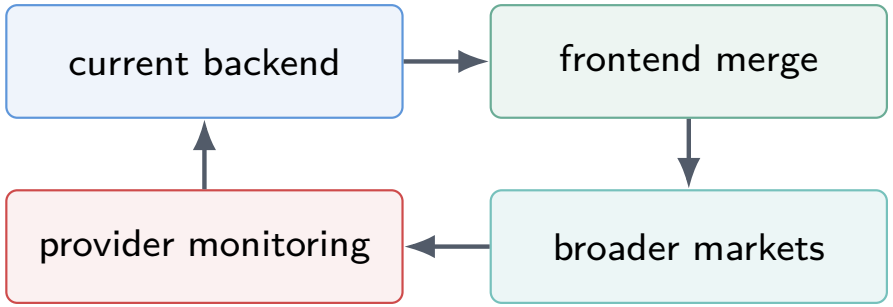


Demo artifacts make the chatbot output traceable

Artifact	Demo highlight
`nlu_result`	product type, question style, entity, risk flags, source plan.
`retrieval_result`	executed sources, ranked documents, structured rows, coverage, warnings.
`analysis_summary`	trend, RSI, valuation, macro direction, and data readiness.
`answer_generation`	evidence IDs, key points, limitations, disclaimer, model name.
`next_question_prediction`	three follow-up questions with scores and reasons.

Future work focuses on integration and robustness

- Merge the chatbot frontend with the backend evidence contract.
- Extend runtime coverage beyond China-market v1.
- Improve live-provider degradation and monitoring.
- Add lightweight document sentiment for low-resource deployment.



FinSight makes the chatbot safer and easier to evaluate

- It separates query understanding, retrieval, analysis, and answer generation.
- It keeps the core path classical, inspectable, and measurable.
- It gives the chatbot traceable JSON instead of ungrounded guesses.
- It supports future UI integration without changing the backend evidence contract.

Thank you!